

Teaching Exam

Subject – Computer Science

Topic – Digital Logic



Lecture No.- 2

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Topics to be Covered

Topic



- 1:- Classification Of Number System ✓
- 2:- All four Weighted Type Number System ✓
- 3:- Binary to Decimal Conversion ✓
- 4:- Decimal To Binary Conversion ✓
- 5:- Binary To decimal Conversion
- 6:- Decimal To octal Conversion ✓
- 7:- Octal to decimal Conversion ✓
- 8:- Decimal To Hexadecimal Conversion ✓
- 9:- Hexadecimal to Decimal Conversion ✓
- 10:- Binary to octal Conversion ✓
- 11:- Binary to Hexadecimal Conversion ✓



Topic : Classification of Number System



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A number is a way to represent arithmetic value, count or measure of a particular quantity. A

number system

can be considered as a mathematical notation of numbers using a set of digits or symbols. In simpler words the number system is a method of representing numbers. Every number system is identified with the help of its base or radix.

Base or Radix of a Number System:

The base or radix of a number system can be referred as the total number of different symbols which can be used in a particular number system. Radix means “root” in Latin. Base equals to 4 implies there are 4 different symbols in that number system. Similarly, base equals to “x” implies there are “x” different symbols in that number system.



Topic : Classification of Number System



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Number System Classification

Positional/ Weighted Number System

- Decimal ✓
- Octal ✓
- Binary ✓
- Hexadecimal ✓
- BCD ✓
- 8-4-2-1 Code ✓

Non-Positional/ Non-Weighted Number System

- Excess-3 Code ✓
- Cyclic Code ✓
- Roma Code ✓
- Gray Code ✓

1. Positional (or Weighted) Number System:

A positional number system is also known as weighted number system. As the name implies there will be a weight associated with each digit. According to its position of occurrence in the number, each digit is weighted.

2. Non-Positional (or Non-weighted) Number System:

Non-positional number system is also known as non-weighted number system. Digit value is independent of its position. Non-positional number system is used for shift position encodes and error detecting purpose.



Topic : Number System /Positional Number System



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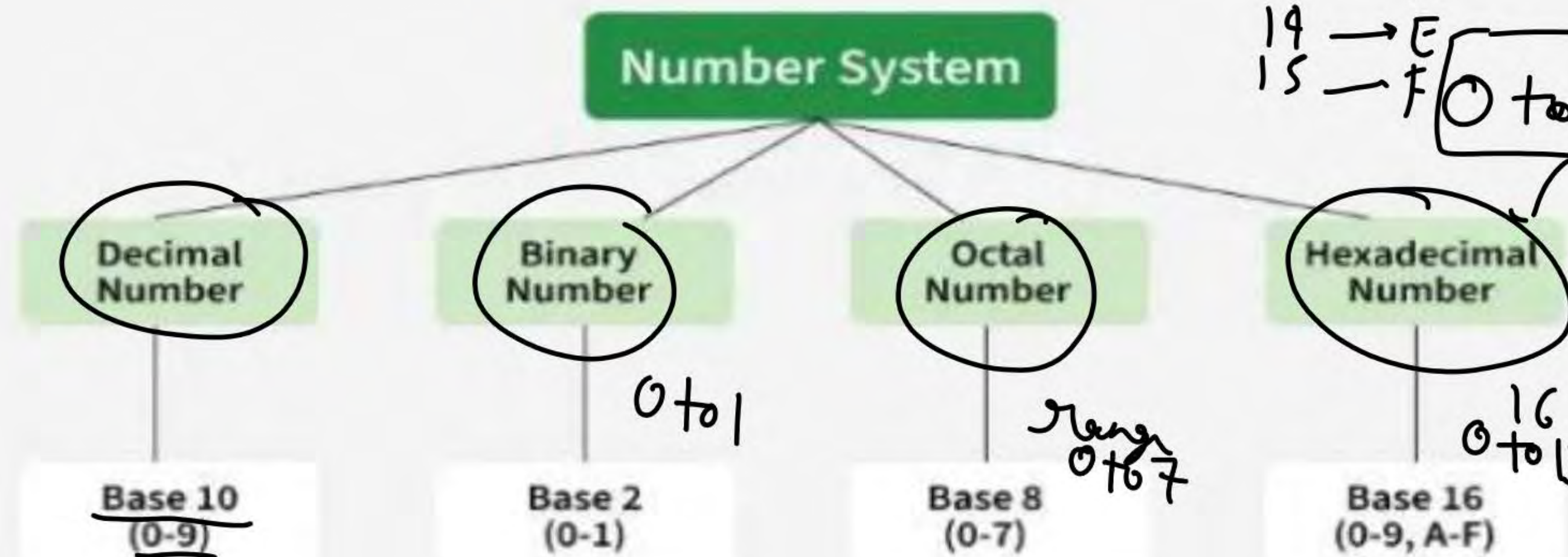
A number system is a way of writing numbers using specific symbols or digits. It helps us represent numbers mathematically. There are different types of number systems, such as the decimal system, binary system, octal system, and hexadecimal system. Here, we will discuss the types of number systems in detail, along with solved examples.

10 → A
11 → B
12 → C
13 → D
14 → E
15 → F

0
:
9

0 to F

0 to 15 but





Topic : Decimal Number System



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Unsigned magnitude (original Binary) $\rightarrow$ Positive $\rightarrow 0$
Signed $\rightarrow$ Negative $\rightarrow 1$

A number system is a method of representing numbers. Every number system is identified with the help of its base or radix. For example, Binary, Octal, Decimal and Hexadecimal Number systems etc. If the **Base value** of a number system is **10**. This is also known as **base-10** or **Decimal number system** which has 10 characters from 0 to 9.

Check whether number is Decimal number system or not? If yes then find base and base value

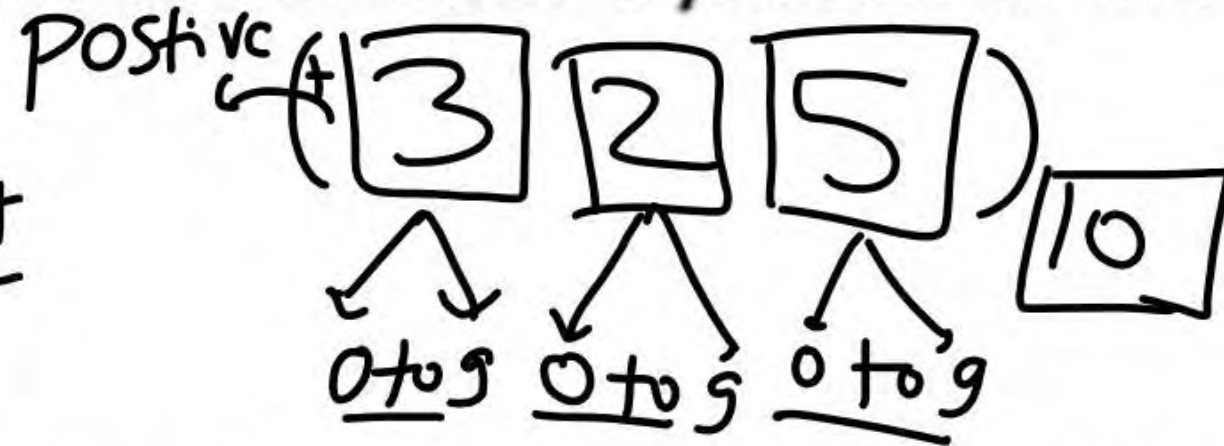
1:- $(325)_{10}$ Correct ✓

2:- $(76.25)_{10}$ Correct ✓

3:- $(-325)_{10}$ Correct (Negative value is possible in Decimal Number System)

4:- $(268)_8$ Incorrect

$-(325)$ ✓





Topic : Binary Number System



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Binary Number System uses two digits, '0' and '1', and is the foundation for all modern computing. The word binary is derived from the word "bi," which means two. But what makes it so essential, and how does it work? This article will dive deep into **binary numbers**, binary decimal number conversion and vice versa, 1's and 2's complements, and how they are used in **computer systems**.

There are generally various types of number systems, and among them, the four major ones are,

- Binary Number System (Number system with Base 2)
- Octal Number System (Number system with Base 8)
- Decimal Number System (Number system with Base 10)
- Hexadecimal Number System (Number system with Base 16)

Base $\rightarrow$ 2
Range 0 and 1

In the Binary Number System, we have a base of 2. The base of the Binary Number System is also called the radix of the number system.

In a binary number system, we represent the number as,

- $(11001)_2$



Check WEATHER Number is Binary or not ?

- 1:- $(1011)_2$ ✓
2:- $(1011)_{10}$ ✓ which is wrong
3:- $(\underline{1011110072})_2$ X
4:- $(1000000123)_2$ X

Each digit check either 0 or 1
Base value = 2



Octal Number System is a number system with base 8 as it uses eight symbols (or digits) namely 0, 1, 2, 3, 4, 5, 6, and 7. For example, 22_8 , 13_8 , 17_8 , etc. are octal numbers. This number system is mainly used in computer programming as it is a compact way of representing binary numbers with each octal number corresponding to three binary digits.

Octal Base Value
= 8
Range 0 to 7

Check whether Number is Octal Number system:-

- 1:- $(625)_8$ ✓
2:- $(78)_8$ ✗
3:- $(795325)_8$ ✗
4:- $(10111)_8$ ✗
- Handwritten notes: $(79)_8$ with a box around 9 and an arrow pointing to 8, and $0 \text{ to } 7$ with an 'X'.



The Hexadecimal Number System is a base-16 number system used in diverse fields, especially in computing and digital electronics. It consists of 16 symbols, including numbers 0 to 9 and letters A to F, offering a compact way to represent binary-coded values. The hexadecimal number system is sometimes also represented as 'hex'.

Hexadecimal = 16
0 to F

Check whether Number is HexaDecimal Number system:-

0 to 9, A, B, C, D, E, F

- 1:- (62FE)₁₆ ✓
- 2:- (FGH)₁₆ ✗
- 3:- (FFFF)₈ ✗
- 4:- (10111)₁₆



How to Write Numbers in Expanded Form?

Go through the below steps to write the numbers in expanded form:

Step 1: Get the standard form of the number.

Step 2: Identify the place value of the given number using the place value chart.

Step 3: Multiply the given digit by its place value and represent the number in the form of (digit $\times$ place value).

Step 4: Finally, represent all the numbers as the sum of (digit $\times$ place value) form, which is the expanded form of the number.



124.
10000.

$$10^0 = 1$$

$$10^n$$

Write down Expanded Form of Decimal Number system

1- (7648)₁₀

2:- (4569.689)₁₀

3:- (999999.00789)₁₀

$$0 \times 10^{-1} + 0 \times 10^{-2} + 7 \times 10^{-3} + 8 \times 10^{-4} + 9 \times 10^{-5}$$

Rule

Before Point (Decimal)

After Point (Decimal)

4 5 6 9

6 8 9

$10^3 \quad 10^2 \quad 10^1 \quad 10^0$

$10^{-1} \quad 10^{-2} \quad 10^{-3}$

Right to Left Index

Left to Right decrease

$$4 \times 10^3 + 5 \times 10^2 + 6 \times 10^1 + 9 \times 10^0 + 6 \times 10^{-1} + 8 \times 10^{-2} + 9 \times 10^{-3}$$

$$4000 + 500 + 60 + 9$$

$$+ 0.6 + 0.08 + 0.009$$

Expanded form

Verify

4569.689

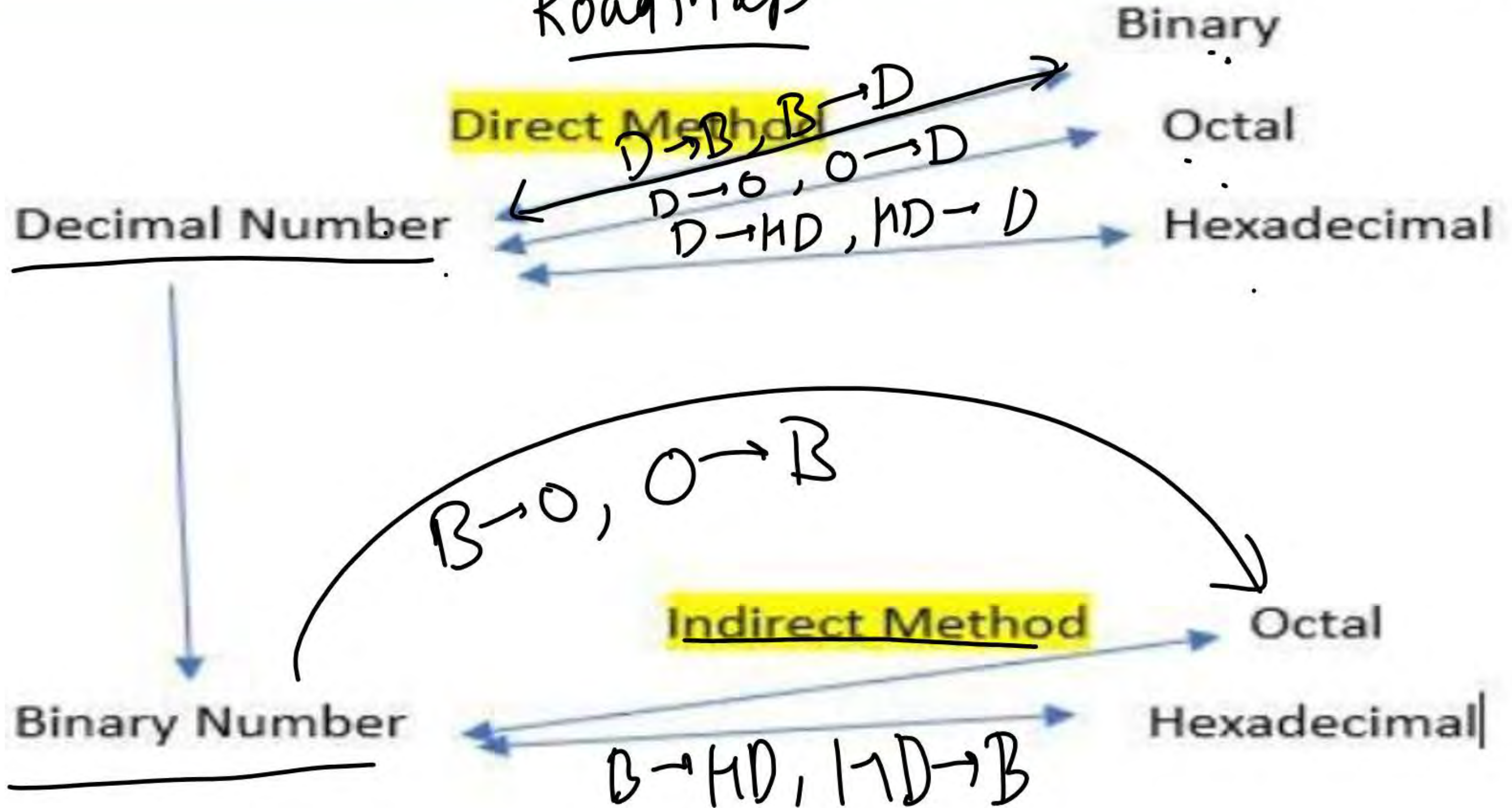
$$10^0 \Rightarrow 1$$

$$99^0 = 1$$

$$100000^0 =$$



Roadmap





How to use Binary to Decimal Calculator?

We can easily use the **binary-to-decimal calculator** by following the steps discussed below,

Step 1: Enter the given value in the binary input field.

Step 2: Click on the convert button to convert the binary value into the decimal value.

Step 3: The value shown as the result is the required value in the decimal form.

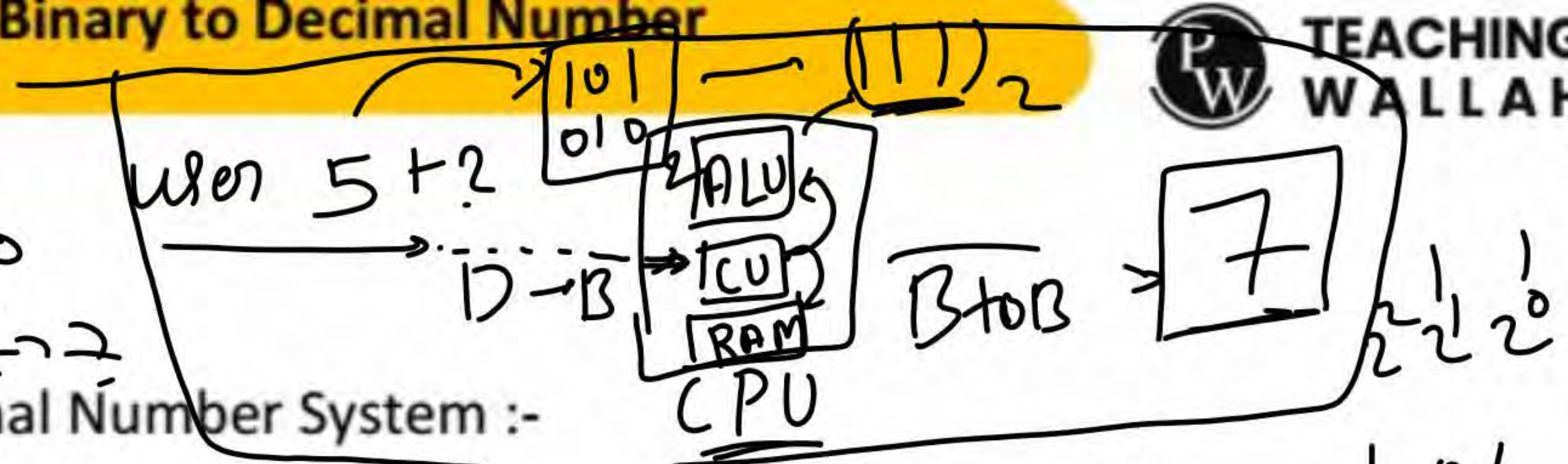
Binary to Decimal Formula

To convert a **binary number to decimal** we need to perform a multiplication operation on each digit of a binary number from right to left with powers of 2 starting from 0 and add each result to get the **decimal** number of it.

Decimal Number = n^{th} bit $\times 2^{n-1}$

Binary to Decimal Formula

$$n = b_n q + b_{n-1} q^{n-2} + \dots + b_2 q^2 + b_1 q^1 + b_0 q^0 + b_{-1} q^{-1} + b_{-2} q^{-2}$$



Convert Binary To Decimal Number System :-

1:- $(10111000)_2 = (?)_{10}$

2:- $(10101010)_2 = (?)_{10} \Rightarrow (170)_{10}$

3:- $(0000100001)_2 = (?)_{10} \Rightarrow (33)_{10}$

0000100001

$2^5 2^4 2^3 2^2 2^1 2^0$

$32 + 1 \Rightarrow 33$

$1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$

$= 128 + 32 + 8 + 2$

$= 170$

101
 010

$2^2 2^1 2^0$



How to Convert Decimal to Binary

To convert decimal numbers into binary, there are several methods, including formulas and division techniques. In this explanation, we'll use the remainder method. The steps to convert a decimal number to binary using this method are as follows:

Step 1: Divide the given decimal number by 2, and find the remainder (R_i).

Step 2: Now divide the quotient (Q_i) that is obtained in the above step by 2, and find the remainder.

Step 3: Repeat the above steps 1 and 2 until 0 is obtained as a quotient.

Step 4: Write down the remainder in the following manner: the last remainder is written first, followed by the rest in reverse order ($R_n, R_{(n-1)}, \dots, R_1$). Thus binary conversion of the given decimal number will be obtained.



$$\begin{array}{r} 2 \overline{) 9999} \rightarrow 1 \\ 2 \overline{) 4999} \rightarrow 1 \\ 2 \overline{) 2499} \rightarrow 1 \\ 2 \overline{) 1249} \rightarrow 1 \\ 2 \overline{) 624} \rightarrow 1 \\ 2 \overline{) 312} \rightarrow 1 \\ 2 \overline{) 156} \rightarrow 1 \\ 2 \overline{) 78} \rightarrow 1 \\ 2 \overline{) 39} \rightarrow 1 \\ 2 \overline{) 19} \rightarrow 1 \\ 2 \overline{) 9} \rightarrow 1 \\ 2 \overline{) 4} \rightarrow 1 \\ 2 \overline{) 2} \rightarrow 1 \\ 2 \overline{) 1} \rightarrow 1 \end{array}$$

Convert Decimal to Binary Number System:

1:- $(789)_{10} = (?)_2$

2:- $(1024)_{10} = (?)_2$

3:- $(9999)_{10} = (?)_2$

$(97.25)_{10}$

$\frac{0}{2^{-1}} \frac{1}{2^{-2}} \frac{0}{2^{-3}}$

$1024 = (1.0000000000000000)_{2^0 2^1 2^2 2^3 2^4 2^5 2^6 2^7 2^8 2^9 2^{10} 2^{11} 2^{12} 2^{13} 2^{14} 2^{15}}$

$4096 = 8000 + 192$

$8) 10(0.125)_{10} = (1011.111)_{2^3 2^2 2^1 2^0 2^{-1} 2^{-2} 2^{-3}}$

$100111000000.125 = (11.875)_{10}$

$\frac{1}{2^2} = \frac{1}{4} = 0.25$



Convert Decimal to Octal with Steps

Follow the steps given below to learn the decimal to octal conversion:

1. Write the given decimal number
2. If the given decimal number is less than 8 the octal number is the same.
3. If the decimal number is greater than 7 then divide the number by 8.
4. Note the remainder, we get after division
5. Repeat step 3 and 4 with the quotient till it is less than 8
6. Now, write the remainders in reverse order (bottom to top)
7. The resultant is the equivalent octal number to the given decimal number.

For example: Convert 1792 into an octal number.

Decimal Number	Operation	Quotient	Remainder	Octal Number
1792	$\div 8$	224	0	0
224	$\div 8$	28	0	00
28	$\div 8$	3	4	400
3	$\div 8$	0	3	3400



Convert $(87.662)_{10}$ to Octal Number

1:- Convert $(127)_{10}$ to Octal.

option

① $(127.25)_8$

8	127
8	15

7
7

120

2:- Convert $(52)_{10}$ to octal.

② $(139.12)_8$ X

③ $(111.25)_8$ X

3:- Convert $(456)_{10}$ to Octal

④ Number 6

$(177)_8$

Result

5
x
456

8	456
8	57

7
(710)₈

16) 506.91 (64)₈

64 + 16 + 7 + 0.25 + 0.412

$(87.662)_{10}$



Octal to Decimal

To convert an octal number to a decimal number we need to multiply each digit of the given octal with the reducing power of 8.

Let us learn here, the conversion of Octal number to Decimal Number or base 8 to base 10.

Examples on Octal to Decimal

Example 1: Suppose 215_8 is an octal number, then its decimal form will be,

$$\begin{aligned}215_8 &= 2 \times 8^2 + 1 \times 8^1 + 5 \times 8^0 \\&= 2 \times 64 + 1 \times 8 + 5 \times 1 = 128 + 8 + 5 \\&= 141_{10}\end{aligned}$$

Example 2: Let 125 is an octal number denoted by 125_8 . Find the decimal number.

$$\begin{aligned}125_8 &= 1 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 \\&= 1 \times 64 + 2 \times 8 + 5 \times 1 = 64 + 16 + 5 \\&= 85_{10}\end{aligned}$$



Decimal Number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Equivalent Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Decimal to Hexadecimal Conversion With Steps

Go through the steps given below to learn how to convert the numbers from decimal to hex.

Step 1: First, divide the decimal number by 16, considering the number as an integer.

Step 2: Keep aside the remainder.

Step 3: Again divide the quotient by 16 and repeat till you get the quotient value equal to zero.

Step 4: Now take the values of the remainder's left in the reverse order to get the hexadecimal numbers.

Note: Remember, from 0 to 9, the numbers will be counted as the same in the decimal system. But from 10 to 15, they are expressed in alphabetical order such as A, B, C, D, E, F and so on.



- 1:- Convert $(960)_{10}$ into hexadecimal.**
- 2:- Convert $(786)_{10}$ into hexadecimal.**
- 3:- Convert $(1616)_{10}$ into hexadecimal.**



As we know, number systems can be converted from one base to another. Thus, we can convert hexadecimal numbers to decimal easily. This **number system conversion** can be done as explained in the example given below:

Example:

Convert 7CF (hex) to decimal.

Solution:

Given hexadecimal number is 7CF.

In hexadecimal system,

$$7 = 7$$

$$C = 12$$

$$F = 15$$

To convert this into a decimal number system, multiply each digit with the powers of 16 starting from units place of the number.

$$7CF = (7 \times 16^2) + (12 \times 16^1) + (15 \times 16^0)$$

$$= (7 \times 256) + (12 \times 16) + (15 \times 1)$$

$$= 1792 + 192 + 15$$

$$= 1999$$



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$$= 1792 + 192 + 15$$

$$= 1999$$



1:- Convert $(1DA6)_{16}$ to decimal system

2:- Convert $(E8B)_{16}$ to decimal system

3:- Convert $(FFF)_{16}$ to decimal system



Using Grouping

Since, there are only 8 digits (from 0 to 7) in octal number system, so we can represent any digit of octal number system using only 3 bit as following below.

Octal Digit Value	Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111



- Take binary number
- Divide the binary digits into groups of three (starting from right) for integer part and start from left for fraction part.
- Convert each group of three binary digits to one octal digit.

This is simple algorithm where you have to grouped binary number and replace their equivalent octal digit.



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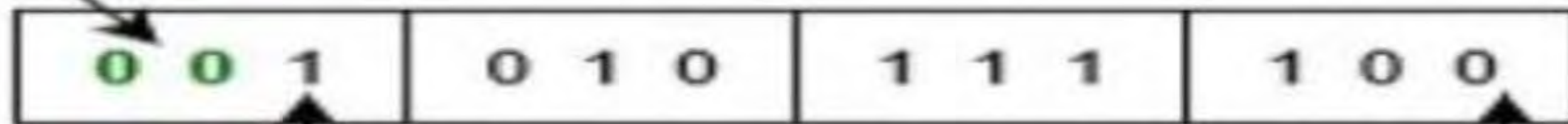
Topic : Binary to Octal Number



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Example-1 – Convert binary number 1010111100 into octal number. Since there is no binary point here and no fractional part. So,

These two 0's are added into MSB to complete group of 3 bits



9th position
(MSB)

0th position
(LSB)

Therefore, Binary to octal is.

$$\begin{aligned} &= (1010111100)_2 \\ &= (001\ 010\ 111\ 100)_2 \\ &= (1\ 2\ 7\ 4)_8 \\ &= (1274)_8 \end{aligned}$$

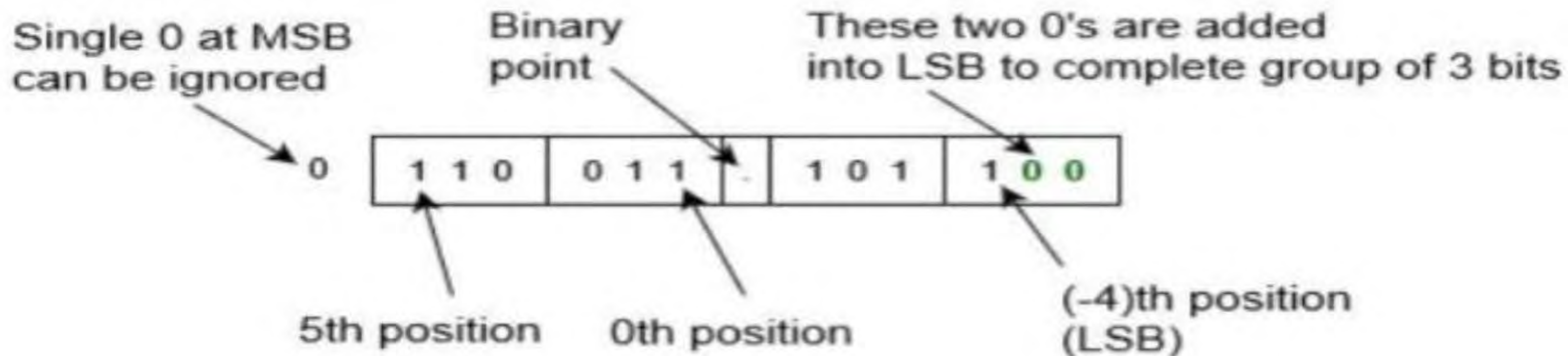


Topic : Binary to Octal Number



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Example-2 Convert binary number 0110 011.1011 into octal number. Since there is binary point here and fractional part. So,



Therefore, Binary to octal is.

$$\begin{aligned} &= (0110\ 011.1011)_2 \\ &= (0\ 110\ 011\ .\ 101\ 1)_2 \\ &= (110\ 011\ .\ 101\ 100)_2 \\ &= (6\ 3\ .\ 5\ 4)_8 \\ &= (63.54)_8 \end{aligned}$$



Using Grouping

Since, there are only 16 digits (from 0 to 7 and A to F) in hexadecimal number system, so we can represent any digit of hexadecimal number system using only 4 bit as following below.

Hexa	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111
Hexa	8	9	A	B	C	D	E	F
Binary	1000	1001	1010	1011	1100	1101	1110	1111

So, if you make each group of 4 bit of binary input number, then replace each group of binary number from its equivalent hexadecimal digits. That will be hexadecimal number of given number. Note that you can add any number of 0's in leftmost bit (or in most significant bit) for integer part and add any number of 0's in rightmost bit (or in least significant bit) for fraction part for completing the group of 4 bit, this does not change value of input binary number.

So, these are following steps to convert a binary number into hexadecimal number.



So, these are following steps to convert a binary number into hexadecimal number.

- Take binary number
- Divide the binary digits into groups of four (starting from right) for integer part and start from left for fraction part.
- Convert each group of four binary digits to one hexadecimal digit.

This is simple algorithm where you have to grouped binary number and replace their equivalent hexadecimal digit.



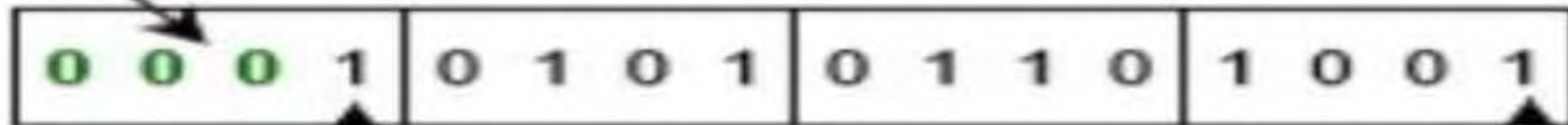
Topic : Binary to HexaDecimal Number



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Example-1 – Convert binary number 1010101101001 into hexadecimal number. Since there is no binary point here and no fractional part. So,

These three 0's are added into MSB to complete group of 4 bits



12th position
(MSB)

0th position
(LSB)

$$= (1010101101001)_2$$

$$= (1\ 0101\ 0110\ 1001)_2$$

$$= (0001\ 0101\ 0110\ 1001)_2$$

$$= (1\ 5\ 6\ 9)_{16}$$

$$= (1569)_{16}$$

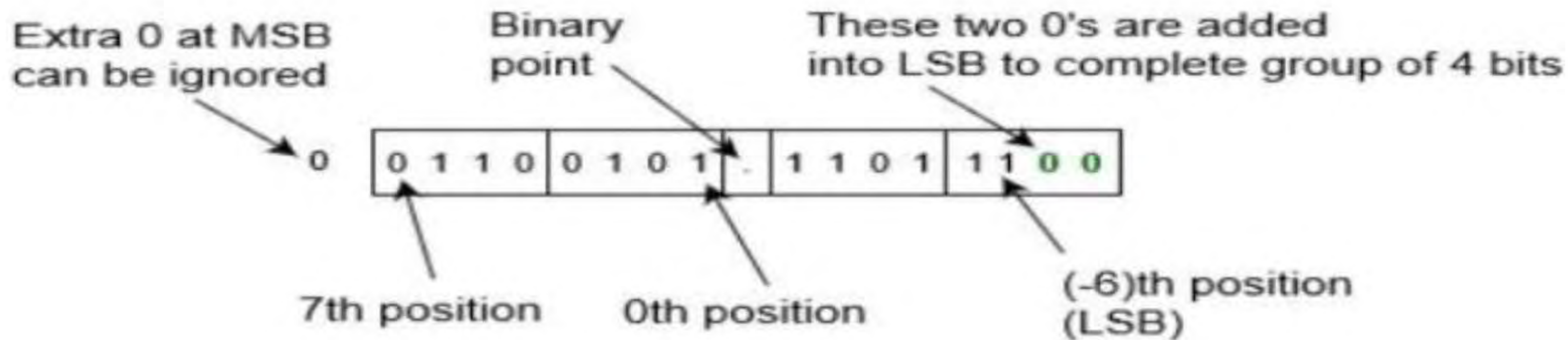


Topic : Binary to HexaDecimal Number



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Example-2 – Convert binary number 001100101.110111 into hexadecimal number. Since there is binary point here and fractional part. So,



Therefore, Binary to hexadecimal is,

$$\begin{aligned} &= (001100101.110111)_2 \\ &= (0\ 0110\ 0101\ .\ 1101\ 1100)_2 \\ &= (0110\ 0101\ .\ 1101\ 1100)_2 \\ &= (6\ 5\ .\ D\ C)_{16} \\ &= (65.DC)_{16} \end{aligned}$$

THANK YOU